ARC-AGI-2 / Qwen3.8-27B - Canonical Project Journal

Experiments, Failures, Lessons, Decisions, Current Progress & Next Work

Updated through the real v2.4G C1 Adaptive SMOKE run - 18 September 2026

Executive purpose

This PDF is the persistent project journal for the ARC-AGI-2 / Qwen3.8-27B solver line. It consolidates the important findings from each major experiment and notebook revision, the errors that were discovered, the lessons retained as regression rules, discussion-derived insights, decisions that proved useful, decisions later corrected, the current verified state, and the exact next experiment.

Evidence discipline. VERIFIED means directly supported by executed notebooks/bundles; INFERENCE means synthesis from multiple verified facts; RETIRED means an assumption or path was invalidated; UNKNOWN means a new experiment is still required. Raw bundles, traces, verifier outputs and hashes remain the authority if this journal and an artifact ever disagree.

Current canonical real-run evidence. The latest decisive run is PIV-Q27-C1-v2.4G, run ID 20260918T104631Z, on 4x NVIDIA L4. The source notebook SHA256 is 50b9acb8b62b3c1860914f562c5154dd58e8906b3342cd8ad6d8875fdc882769; the executed notebook SHA256 is 44934b207807f2b4a870304205b579f78daea3f23b64720aa6fec73455ffb66f; the evidence bundle SHA256 is 387d4bb7383c9e7917c6deff5d2999e0ab83ff9b105d1487aa7dcd7a8cb56837. The bundle contains 82 members: 81 manifest-listed evidence files plus MANIFEST.csv; all 81 manifest entries passed size/SHA verification.

Purpose. This is the persistent experiment/decision log for the project. It records the important findings, failures, corrections, lessons, discussion-derived insights, decisions that proved useful, decisions later revised, current evidence state, and the exact next experiment.
Evidence labels: VERIFIED = directly supported by executed artifacts/bundles; **INFERENCE** = synthesis from multiple verified facts; **RETIRED** = invalidated/obsolete assumption; **UNKNOWN** = requires a new experiment.
Do not use this journal as a substitute for raw evidence. Raw bundles, traces, verifier outputs and hashes remain the authority.

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0. Current executive snapshot

Current state: mechanism works; semantic program quality is now the primary blocker.

The v2.4G real 4xL4 run is the first run in this lineage where the complete agent pipeline reached a standalone program on **every real trajectory**:

- **16/16 final programs produced**
- **16/16 syntax-valid**
- **16/16 fresh verifier executions**
- **16/16 terminal JSON artifacts parsed**
- **0 infrastructure failures**
- **0 budget deaths**
- **0/16 all-train-exact**

This moves the bottleneck from **artifact production/finalization** to **program correctness**.

Canonical v2.4G real-run identity

Item	Value
Run ID	20260918T104631Z
Profile	ADAPTIVE_SMOKE
GPU	4x NVIDIA L4
Canonical source SHA256	50b9acb8b62b3c1860914f562c5154dd58e8906b3342cd8ad6d8875fdc882769
Executed notebook SHA256	44934b207807f2b4a870304205b579f78daea3f23b64720aa6fec73455ffb66f
Evidence bundle SHA256	387d4bb7383c9e7917c6deff5d2999e0ab83ff9b105d1487aa7dcd7a8cb56837
Bundle manifest	81/81 entries verified; 0 SHA/size mismatches
Code source match	Executed code cells are source-exact to canonical v2.4G
Total elapsed	4691.31 sec ~ 78.19 min
Model load	8.25 min
Warmup	0.17 min

Real arm results

Metric	ACT-LOW	ACT-BUDGET3072
Tool engagement	75.0%	100.0%
Final program	100.0%	100.0%
Syntax valid	100.0%	100.0%
Verifier executed	100.0%	100.0%
All-train-exact	0.0%	0.0%
Budget death	0.0%	0.0%
Infrastructure failure	0.0%	0.0%
Mean generated tokens/task	9180	9371
sec/output-token	0.0464	0.0513
Proxy48 projected +20%	4.75 h	3.64 h

Most important near-quality signal: ACT-LOW / `291dc1e1` matched **2/4 training examples exactly**. This is the strongest current seed for verifier-feedback repair. BUDGET3072 had zero exact train pairs across all eight tasks; `136b0064` timed out in the train verifier.

1. Experiment chronology - finding -> failure -> fix -> lasting lesson

A. Direct/Q1 baseline

Direct OFF vs XHIGH

- **VERIFIED:** direct generation did not solve tasks; OFF produced some valid grids but no exact task; XHIGH frequently spent the whole generation budget before useful action.
- **Lesson:** raw model generation is insufficient; code/tool interaction is needed.
- **Persistent rule:** use fresh program verification, not prose/grid plausibility, as correctness authority.

B. PIV harness lineage v1.0 -> v1.7

v1.0 - first stateful REPL + fresh verifier

- Introduced the right core direction: persistent Python, puzzle preloaded/protected, final `solution(grid)`, fresh verification.
- Early implementation still had source/runtime mismatches.

v1.2 - separate train-perfect from later test execution

- **Bug fixed:** a test-stage failure could erase valid train-perfect evidence.
- **Persistent rule:** train correctness and test execution are separate states.

v1.3 - parser/retry/grid/worker/sandbox hardening

- Added bounded retry, think-aware parsing, worker restart, tolerant grid normalization, partial-output preservation.
- **Lesson:** compile-clean code does not imply runtime-safe agent behavior.

v1.4 - same-history retry vs outer restart

- **Bug fixed:** local retry and outer source-style restart had been conflated.
- **Persistent rule:** preserve REPL/history on step retry; use outer restart only after bounded retry exhaustion.

v1.5 - viability gate correction

- Replaced weak first-turn-length heuristic with terminal budget-death + empirical wall-clock feasibility.
- **Lesson:** measure the terminal failure mode, not a proxy that can be misleading.

v1.6 - deadline/SMOKE8/accounting hardening

- Added projected-deadline protection, representative stratified SMOKE8, dynamic guard accounting, sanitized tool feedback, terminal coverage and real-scheduler PREFLIGHT.
- **Lesson:** harness invariants must be actively exercised, not merely present.

v1.7 - dynamic multi-call budget + canonical XHIGH failure

- **VERIFIED real GPU:** 8/8 tasks consumed 12,288 tokens in first-turn thinking; tool engagement 0/8; final programs 0/8; all-train-exact 0/8; infrastructure clean.

- **Root cause:** template/reasoning-budget interaction, not REPL/parser/verifier failure.
- **Persistent rule:** XHIGH natural reasoning is a retired path under this budget.

C. Reasoning activation v1.8 -> v1.9d

LOW / MEDIUM / BUDGET3072 probe

- **VERIFIED real GPU:** LOW 7/8 tool activation; BUDGET3072 7/8; MEDIUM 1/8; XHIGH frozen reference 0/8.
- LOW tool execution was materially cleaner and cheaper than BUDGET3072 in the cheap canary.
- **Decision:** retain LOW and BUDGET3072 independently; retire MEDIUM from near-term solver runs.
- **Correction:** `reasoning_effort="high"` is invalid for this model family; only `xhigh/medium/low`.
- **Persistent rule:** activation is not solving. Probe zeros do not establish solver-quality failure.

Forced-close lessons

- Forced close only exits thinking; it does **not** guarantee a tool call or final code.
- Token-prefix continuation must remain exact; no silent text re-render.
- Guard rejection must not consume executed tool budget.
- Phase-2 seed remains separate and deterministic.

D. Throughput v1.2.4

C1 graph/noAR

- **VERIFIED real 4xL4:** C0 long@8 ~ 48.30 tok/s; C1 graph/noAR ~ 129.17 tok/s; ~2.67x speedup.
- Custom all-reduce added no meaningful gain.
- C1 became the prime serving candidate.

Retired numerics gate

- The old `greedy-prefix agreement >= 0.50` safety gate was invalidated because the unchanged C0 repeat itself scored ~0.1875.
- **RETIRED:** token fingerprint/prefix agreement as solver-correctness authority.
- **Persistent rule:** serving safety must be judged by actual solver behavior: final program, fresh all-train-exact, test execution, official exact.

Prefix64 / TP2 OOM

- **Lesson:** stressed OOM rows are capacity failures, not proof that a feature is intrinsically unsupported.
- Parent-side fail-fast was insufficient and wasted large wall time.
- **Persistent rule:** terminal worker failure should be reclaimed quickly; do not wait full timeout if failure is already known.

E. P1-SPARSE side branch

- Aggregate outcomes were interesting but game-level variance and teardown/runtime-health issues remained.

- **Decision:** PROMISING/HOLD; keep its metric logic separate from the Qwen ARC-AGI-2 main solver.
- **Persistent rule:** side-branch scores cannot drive main-line promotion.

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2. C1 solver-quality lineage v2.0 -> v2.3

v2.0 -> v2.2

Correctness/failure-path audits found and fixed: - train-exact promotion without usable test output; - strict task metric mislabeling; - verifier state overstatement; - test timeout erasing train evidence; - generate exception handling; - unsafe phase-2 draft salvage; - infrastructure failure being confused with model-quality failure.

v2.3 PREFLIGHT

- Harness cleared the major P0/P1 audit issues.
- Important rule locked: **PREFLIGHT is harness evidence, not model success.**

v2.3 real C1 quality smoke

VERIFIED real GPU: - C1 serving worked. - LOW tool engagement 7/8; BUDGET3072 8/8. - Persistent REPL was genuinely used. - But **0 final programs / 16 trajectories**, therefore verifier was never reached. - LOW spent ~92% of generation in thinking; BUDGET forced close frequently led to another tool call/malformed action rather than a final program. - **Finding:** bottleneck moved from “can the model act?” to “can it commit a standalone program?” - **Decision:** do not run Proxy48/Public. Build a protected terminal commit mechanism.

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3. Terminal-commit lineage v2.4 -> v2.4G

v2.4 - protected terminal budget

Design: - total 12,288 tokens/task unchanged; - exploration <=8,192; - protected final reserve 4,096; - non-thinking, no-tools structured finalizer; - fresh verifier remains authority.

Strategic correction

The first roadmap was too serial: PREFLIGHT -> upload/review -> 2-task pilot -> upload/review -> SMOKE8. We changed to **result-driven one-session execution**: embedded safety checks -> positive control -> 2-task interface diagnostic -> automatic SMOKE8 if structurally clean.

Lesson: after basic infrastructure is stable, endless review loops can become the bottleneck themselves.

v2.4A

Found deterministic blockers: - terminal wave `"phase":"terminal_finalizer"` crashed downstream `int(phase)`; - adaptive gate could false-pass a broken finalizer and false-fail a natural early solution; - runtime accumulation was overwritten; - context-reserve guarantee was incomplete.

v2.4B

Major fixes: - hardened structured JSON parsing and special-token handling; - safe phase coercion; - artifact-based adaptive interface gate; - cumulative stage runtime; - dedented tool payloads; - verifier return contract consistency; - engine output-count guards.

v2.4C

Fixed: - verifier infrastructure propagation; - terminal context reserve against tool-feedback growth; - full-SMOKE-only recommendation scope.

v2.4D

Fixed: - bounded reasoning handoff to finalizer; - retry exhaustion now uses protected terminal slot; - REPL startup/runtime infrastructure propagation; - unclosed-thinking draft not treated as committed final; - terminal JSON parser no longer mutates control-token-looking literals inside program strings; - positive control includes fresh verifier; - Proxy readiness respects compute/promotion gate.

v2.4E

Fixed two runtime blockers: - parser self-test had stale expectations around unclosed thinking; - PREFLIGHT recommendation variable initialization.

v2.4F

Fixed: - PREFLIGHT terminal-policy assertion; - shared terminal-trigger matrix; - engine output cap guard; - retry-accounting noise at terminal boundary; - TERMINAL_PILOT report labels.

v2.4G

Fixed defensive cap-guard blast radius: - a single per-request `max_tokens` overrun no longer kills the whole wave/arm; - output-count mismatch remains wave-fatal because mapping becomes untrustworthy.

Hard-stop decision before run: no more pre-run micro-patching unless a deterministic default-path blocker is reproduced. The next information had to come from real GPU execution.

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4. v2.4G real GPU run - what it actually proved

4.1 Integrity

- **VERIFIED:** executed code cells are source-exact to canonical v2.4G.
- **VERIFIED:** evidence bundle manifest 81/81 entries passed SHA/size verification.
- **VERIFIED:** engine bootstrap complete on 4xL4; C1 engine fingerprint unchanged.
- **VERIFIED:** real-Qwen terminal positive control passed structured output -> parser -> fresh verifier.
- **VERIFIED:** adaptive 2-task gate passed; remaining SMOKE6 ran automatically.

4.2 The central progress result

Before v2.4G (v2.3)

reasoning -> tools -> no final artifact

Now

reasoning -> tools -> terminal commit -> standalone code -> fresh verifier -> wrong program

This is **substantial progress**, because harness/interface failure is no longer the dominant blocker.

4.3 ACT-LOW

- Tool engagement: 75%.
- 17/17 executed tool calls succeeded; no exceptions/timeouts/guard rejects.
- 8/8 final programs; 8/8 syntax valid; 8/8 verifier executed.
- 0/8 all-train-exact.
- Train pair evidence: `291dc1e1` achieved 2/4 exact training pairs; all other LOW tasks 0 exact pairs.
- Finalizer artifact parse failures: 0.
- Mean generated tokens/task: ~9,180.
- Median unused token budget after candidate #1: ~2646; worst-case unused budget across LOW tasks: ~1229.
- Runtime: pilot 2 ~ 9.58 min; remaining 6 ~ 28.46 min; total LOW loop ~ 38.04 min.

4.4 ACT-BUDGET3072

- Tool engagement: 100%.
- 22 executed calls: 19 OK, 3 Python exceptions; no tool timeout/guard rejection.
- 8/8 final programs; 8/8 syntax valid; 8/8 verifier executed.
- 0/8 all-train-exact.
- 7 train_rejected; `136b0064` train verifier timeout.
- 17 forced-close turns; 5 action-cap hits; BUDGET still operates in roughly two full forced-close turns + partial third under 8,192 exploration.
- Median unused token budget after candidate #1: ~2758; worst-case unused budget across BUDGET tasks: ~2175.
- Runtime: pilot 2 ~ 11.13 min; remaining 6 ~ 17.67 min; total BUDGET loop ~ 28.80 min.

4.5 What we must NOT conclude

- Do **not** declare LOW or BUDGET a quality winner: correctness is 0/8 for both.
- Do **not** run Proxy48 merely because compute gate passed.
- Do **not** return to terminal-reserve tuning: terminal artifact production succeeded 16/16.
- Do **not** infer that C1 hurts quality from this run; there is no matched C0-quality comparator here and the current blocker is semantic synthesis.
- Do **not** call BUDGET “bad” solely because it has fewer effective exploration turns; that is a known property of the chosen terminal-policy package.

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5. Decisions that proved useful

1. **Keeping LOW and BUDGET3072 independent** after activation. LOW failing would not have been a valid reason to suppress BUDGET, and vice versa.
2. **Retiring MEDIUM** from near-term quality experiments after weak activation.
3. **Using only one serving intervention:** `enforce_eager=True -> False`; custom AR and prefix cache stayed off.
4. **Rejecting the old greedy-prefix numerics gate.**
5. **Blocking Proxy48/Public until real verified single-lineage quality exists.**
6. **Preserving fresh all-train-exact as the correctness authority.**
7. **Separating infrastructure validity from model-quality failure.**
8. **Insisting on source hashes, executed notebook, bundle manifest and trace evidence.**
9. **Switching from review-driven to result-driven execution** once the harness became stable.
10. **Freezing v2.4G and actually running it** instead of continuing pre-run patch loops.

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6. Decisions/assumptions later corrected

1. **“Compile/nbformat pass means runtime-safe.” - RETIRED.** Sequential execution and targeted counterexamples are necessary.
2. **“PREFLIGHT pass means model will activate/solve.” - RETIRED.** Fake-engine success proves harness behavior only.
3. **“Forced close guarantees a tool/final answer.” - RETIRED.**
4. **“Tool engagement implies solving progress.” - RETIRED as a promotion rule.** It is diagnostic only.
5. **“Greedy-prefix >=0.50 proves serving safety.” - RETIRED.**
6. **“OOM means prefix/TP feature unsupported.” - RETIRED.** It may be capacity/concurrency stress.
7. **“Every notebook revision deserves a separate upload/review milestone.” - REVISED.** This created an avoidable serial loop.
8. **Unsupported attribution that Tufa Labs used Qwen3.8-27B on ARC-AGI-2 - CORRECTED.** Public evidence supported a Qwen-27B-class ARC-AGI-3 harness, not that specific ARC-AGI-2 model claim.
9. **Early tendency to keep polishing the harness after v2.3 - REVISED.** The project imposed a hard stop and required a real GPU result.
10. **Runtime estimate 2.5-3.5 h for v2.4G - OVERESTIMATED.** Actual full run was ~78.2 min.

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7. Regression firewall - rules that must persist

- No `reasoning_effort="high"`.
- Preserve exact forced-close/token-prefix continuation semantics; never silently text-rerender phase 2.
- Guard rejection does not consume executed tool budget.
- Same-history retry and outer restart remain distinct.
- Train-perfect evidence cannot be erased by later test failure.
- Missing/duplicate terminal task IDs fail coverage.

- Infrastructure failure is separate from candidate correctness.
- No old greedy-prefix/raw-cell similarity quality gate.
- TP2/prefix OOM is not feature-infeasibility proof.
- No prompt search/PTQ/NVARC/reviewer/evolution/seed diversification mixed into a single-mechanism experiment.
- No Proxy48/Public until at least one reproducible fresh-verifier all-train-exact program with usable test execution.
- One major mechanism per experiment.
- New P2/P3 issues are logged/deferred; only reproduced P0/P1 blockers should interrupt the run cadence.

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8. Current progress state

VERIFIED

- Stateful REPL/tool transport works.
- LOW and BUDGET3072 activate tools on real Qwen.
- C1 graph/noAR serving is fast enough for SMOKE and projected Proxy48 compute.
- Terminal reserve/finalizer solves the previous artifact-production bottleneck.
- Structured terminal output, parser, context reserve, fresh verifier and adaptive gate all worked in the real run.
- Program syntax/execution validity is no longer the immediate blocker.

UNKNOWN / unresolved

- Can verifier feedback repair a partially correct candidate into all-train-exact?
- Which arm is better after repair, not merely before repair?
- Can single-lineage quality become reproducibly non-zero?
- Once non-zero quality exists, how much does multi-sample/pass@K help?
- Does a matched C0-LOW quality control materially differ from C1? Only needed if serving behavior becomes suspect.
- Prefix caching/TP2 remain later throughput questions, not current solver questions.

RETIRED / do not reopen now

- XHIGH natural full-budget path.
- MEDIUM near-term solver arm.
- Old numerics/prefix correctness gate.
- Terminal reserve tuning as the primary next step.
- Broad pre-run notebook review loops without an observed blocker.

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9. Exact next experiment - v2.5 Candidate-First Verifier-Feedback Repair

Primary question: after candidate #1 fails fresh train verification, can one compact verifier-feedback repair turn produce an all-train-exact program?

Frozen from v2.4G

- same model/runtime/C1 engine;
- same frozen SMOKE8 tasks/order;
- same two arms and seed policy;
- same REPL, terminal commit, JSON parser and fresh verifier;
- same 12,288 total token ceiling;
- same test embargo: no test execution until all training pairs are exact.

Single new mechanism

For a failed candidate: 1. run fresh train verifier; 2. construct compact feedback containing only training-side evidence: failed training indices, execution error if any, and bounded candidate-prediction vs expected-grid diagnostics; 3. verify that the feedback is actually present in the repair prompt using hash + exact-presence instrumentation; 4. make **one** non-thinking/no-tool structured repair call; 5. fresh-verify candidate #2; 6. only if all-train-exact, execute tests.

Why one repair round first

- v2.4G leaves median unused budget of about **2646 LOW tokens** and **2758 BUDGET tokens** after candidate #1.
- Every real task retained at least roughly **1229 tokens** of unused total budget.
- Therefore repair can be tested inside the same 12,288-token cap without introducing a larger-budget confound.

Diagnostic anchor

`291dc1e1` / ACT-LOW is the strongest current seed because candidate #1 was already **2/4 train-exact**.

Run discipline

Do **not** return to a 2-task -> upload -> review loop. Use one notebook, full frozen SMOKE8, same two arms, one repair round, then inspect real quality evidence.

Promotion gate after v2.5

- `all_train_exact > 0` + usable test execution -> reproduce once, then Proxy48.
- final programs remain 100% but repaired train-exact remains 0 -> inspect semantic failure taxonomy / candidate-search strategy next.
- infrastructure invalid -> fix only the observed blocker.
- do not add multi-sample, reviewer, DSL or evolution until single-lineage repair quality is non-zero.

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10. Canonical artifact map

Artifact	Role
ARC_AGI2_Qwen38_Deep_Audit_Reference_2026-09-17.pdf	pre-v2.4 canonical historical audit
PIV-Q27-C1-v2_4G-ONE-SESSION-ADAPTIVE-SMOKE-FINAL.ipynb	canonical source for decisive terminal-commit run
experiment-1-arc-agi-2-with-qwen-3-8 (12)(5).ipynb	executed v2.4G notebook
PIV_Q27_C1_V24G_ADAPTIVE_SMOKE_BUNDLE.zip	canonical v2.4G evidence
This annotated notebook	living project journal + executed evidence notebook; **markdown-only annotation, solver code unchanged**

Final status line

Progress is real: the project has moved from **no action -> tools -> no final artifact -> 100% executable final artifacts**.
The next bottleneck is now narrowly defined: semantic program repair from fresh verifier feedback.
The next experiment should produce new solver-quality evidence, not another harness-hardening loop.

Appendix A - Compact experiment ledger

Stage / version	Real status	Main finding / failure	Lasting lesson / action
Direct baseline OFF	Real generation	Some valid grids, no exact solve	Direct output is not enough; use executable programs + verification
Direct baseline XHIGH	Real generation	Reasoning consumed budget before useful action	Natural XHIGH under this budget is a retired path
PIV v1.0	Harness	Correct stateful-REPL direction, early mismatches remained	Preserve persistent REPL + fresh verifier semantics
PIV v1.2	Harness	Test-stage failure could erase train-perfect evidence	Train correctness and test execution are separate states
PIV v1.3	Harness	Parser/retry/worker/grid edge cases	Compile success is not runtime safety
PIV v1.4	Harness	Same-history retry vs outer restart corrected	Keep the two retry levels distinct
PIV v1.5	Harness	Weak first-turn gate replaced	Gate on real terminal budget death + wall time
PIV v1.6	Harness	Deadline, SMOKE8 composition, guard accounting, sanitization	Exercise invariants in real scheduler tests
PIV v1.7 XHIGH SMOKE	Real GPU	8/8 first-turn 12,288-token deaths; 0 tools/finals	Template-budget interaction was the dominant failure
ACT v1.8/v1.9d	Real GPU probe	LOW 7/8 tools; BUDGET3072 7/8; MEDIUM 1/8	Keep LOW + BUDGET independently; retire MEDIUM near-term

Stage / version	Real status	Main finding / failure	Lasting lesson / action
Throughput C0	Real GPU	Long@8 ~48.30 tok/s	Baseline serving reference
Throughput C1 graph/noAR	Real GPU	Long@8 ~129.17 tok/s, ~2.67x	Prime serving candidate
Throughput custom AR	Real GPU	No meaningful gain over C1	Keep custom AR disabled
Old greedy-prefix gate	Audit	Unchanged C0 repeat also failed threshold	RETIRED as quality/safety authority
Prefix64 / TP2 stress	Real GPU	OOM / watchdog under aggressive capacity	Capacity failure != feature infeasibility
P1-SPARSE A/B	Real GPU side branch	Interesting aggregate, game-level variance/runtime issues	PROMISING/HOLD; do not mix score logic with main line
C1 v2.0	Harness revision	Promotion/state labeling issues found	Quality requires fresh train + usable test evidence
C1 v2.1	Harness revision	Verifier state, test timeout, generation handling hardened	Preserve evidence through failure paths
C1 v2.2	Harness revision	Unsafe phase-2 salvage / infra interpretation found	Draft reasoning is not final; infra != model quality
C1 v2.3 PREFLIGHT	CPU/harness	Major blockers cleared	PREFLIGHT proves harness only
C1 v2.3 quality SMOKE	Real GPU	Tools active, REPL works, 0/16 final programs	New blocker: terminal commitment
v2.4 terminal design	Solver-policy revision	Protected 4,096-token terminal reserve	Do not let exploration consume final artifact budget
v2.4A	Audit	Deterministic phase crash, bad adaptive gate, runtime overwrite	One-session workflow needed robust terminal accounting
v2.4B	Audit/revision	JSON/control-token, gate, runtime, payload fixes	Terminal artifact path must be tested end-to-end
v2.4C	Audit/revision	Verifier infra + hard context reserve	Protect both token and context capacity
v2.4D	Audit/revision	Reasoning handoff, retry-finalizer, REPL infra, parser boundary	Commit boundary and infrastructure propagation matter
v2.4E	Audit/revision	Two pre-run runtime blockers fixed	Targeted runtime counterexamples beat syntax-only checks
v2.4F	Audit/revision	PREFLIGHT terminal policy, trigger matrix, cap guard	Shared production/self-test trigger logic reduces drift
v2.4G	Audit + real GPU	16/16 executable final programs, 0/16 all-train-exact	Artifact bottleneck solved; semantic program correctness is next

Appendix B - Decision log: what to keep and what not to repeat

Decisions that were correct and should persist

- Preserve LOW and BUDGET3072 as independent arms after activation; never gate one on the other.

- Keep the serving intervention minimal: enforce_eager=True -> False; custom AR and prefix cache remain off.
- Use fresh all-train-exact verification as the correctness authority.
- Block Proxy48/Public scaling until single-lineage verified quality is non-zero and reproducible.
- Separate infrastructure validity from model-quality failure.
- Require source SHA, executed notebook, evidence bundle, manifest and raw traces for decisive claims.
- Freeze a working notebook and run it once the remaining issues are P2/P3 rather than continuing review-driven micro-patching.

Decisions / assumptions that were corrected

- Compile/nbformat success is not enough for runtime safety.
- PREFLIGHT/fake-engine success is not real-model success.
- Forced close does not guarantee a tool call or final program.
- Tool engagement is diagnostic, not a promotion metric.
- Greedy-prefix similarity is not a solver-correctness gate.
- OOM under stressed TP/prefix settings does not prove the feature is unsupported.
- A separate upload/review milestone for every notebook version became counterproductive once the harness stabilized.
- The earlier attribution that Tufa Labs was known to use Qwen3.8-27B for ARC-AGI-2 was unsupported and was corrected; public evidence only supported a related Qwen-27B-class ARC-AGI-3 harness.
- Pre-run runtime estimates above two hours for v2.4G were too pessimistic; the measured full run was ~78.19 minutes.

Appendix C - Current go/no-go rules

1. **Next mechanism:** one candidate-first verifier-feedback repair round, not more terminal-reserve tuning.
2. **Proxy48 gate:** at least one reproducible fresh-verifier all-train-exact program plus usable test execution.
3. **If repair still yields 0 train-exact:** inspect semantic failure taxonomy/candidate-search strategy; do not immediately add pass@K/reviewer/evolution.
4. **If infrastructure breaks:** fix only the observed blocker.
5. **Do not reopen retired paths** (natural XHIGH, MEDIUM near-term, old numerics gate, broad pre-run review loops) without new evidence.

Appendix D - Evidence hierarchy and canonical artifacts

1. PIV_Q27_C1_V24G_ADAPTIVE_SMOKE_BUNDLE.zip - current real-run evidence authority.
2. experiment-1-arc-agi-2-with-qwen-3-8 (12)(5).ipynb - executed v2.4G notebook.
3. PIV-Q27-C1-v2_4G-ONE-SESSION-ADAPTIVE-SMOKE-FINAL.ipynb - canonical v2.4G source.
4. ARC_AGI2_Qwen38_Deep_Audit_Reference_2026-09-17.pdf - canonical historical audit before the terminal-commit run.
5. This journal - decision/learning synthesis; raw artifacts override it if any discrepancy is found.